Supplementary Material

Abstract

This document serves as supplementary material for the article *"Radiation Pattern Design for SINR Maximization in Multi-User Reconfigurable MISO Communications"*. It is divided into two parts: the first part provides a more detailed description in Fig. 2 of the article, while the second part summarizes the abbreviations used throughout the article to enhance clarity for the readers.

I. Radiation Pattern of RAs

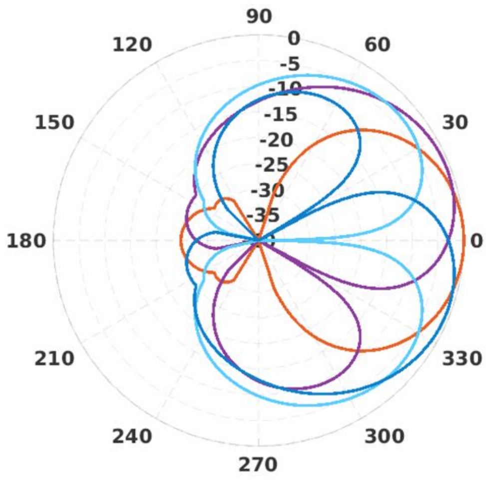

Fig.1 (a) Horizontal radiation pattern

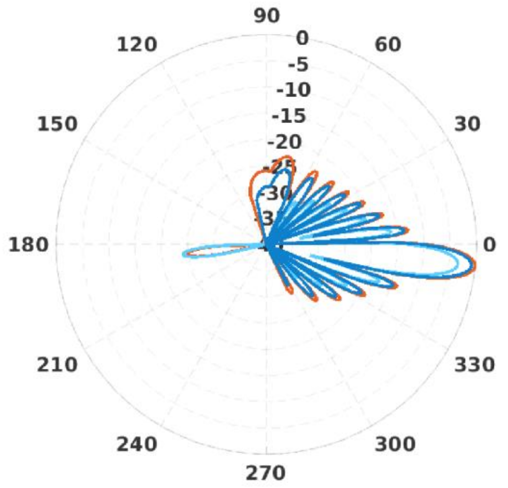

Fig.1(b) Vertical radiation pattern

Mode 1 Mode 2 Mode 3 Mode 4

Figure.1(a) and Figure.1(b) show the horizontal and vertical radiation patterns produced by RAs, respectively. Denoting the azimuth angle with respect to the local coordinate of the BS by ϕ, we consider four pattern mode of a RA:

1、Mode 1: Normal pattern with 51° beamwidth and peak gain at -1°;

2、Mode 2: Tilt pattern with 51° beamwidth and peak gain at 20°;

3、Mode 3: Tilt pattern with 51° beamwidth and peak gain at -20°;

4、Mode 4: Split pattern with 51° beamwidth.

Intuitively, traditional antennas exhibit a fixed radiation pattern orientation, as exemplified by Mode 1, which results in relatively lower coverage gain compared to RAs capable of offering multiple pattern options. Notably, practical RAs can generate a variety of radiation pattern shapes, such as the split pattern demonstrated in Mode 4. This capability enables RAs to provide a significantly broader range of patterns than those achievable by merely rotating conventional antenna patterns. Moreover, four distinct fundamental reconfigurable radiation patterns have been validated as physically realizable, with the corresponding data available in the folder named "data" under the same link.

II. Abbreviation of the Terms

As shown in Table I, we have prepared a table summarizing the abbreviations, to assist readers in better understand the terms of the paper.

TABLE I: Abbreviation of the terms in the paper

Abbreviation	Definition
SISO	Single-input single-output
MIMO	Multiple-input multiple-output
MU-MISO	Multi-user multiple-input multiple-output
RA	Reconfigurable antenna
PR-MIMO	Pattern reconfigurable multiple-input multiple-output
PR-MU-MISO	Pattern reconfigurable Multi-user multiple-input multiple-output
CSI	Channel state information
QoS	Quality of service
DPC	Dirty paper coding
THP	Tomlinson-Harashima precoding
VP	Vector perturbation
ZF	Zero-forcing
RZF	Regularized zero-forcing
PM	Power minimization
SINR	Signal-to-interference-plus-noise-ratio
SDP	Semidefinite programming
DoF	Degree of freedom
SSK	Space-shift-keying
SM	Space modulation
AO	Alternating optimization
SVO	Singular value optimization
SCA	Successive convex approximation
SER	Symbol error rate
AoD	Azimuth angle of departure
EAoD	Elevation angle of departure
FP	Fractional programming
KKT	Karush-kuhn-tucker